

Fault Location Method for Distribution Network Based on Improved Apparent Impedance

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Abstract—The safety and stability of distribution network is directly related to the reliability of costumers. However, the complexity of the structure and variety of conductors in distribution network makes the fault location methods based on uniformed parameters unsuitable for distribution network. Therefore, an improved apparent impedance fault location method based binary search is proposed in this paper. In order to avoid calculated the fault distance directly, the method obtain the fault position by comparing difference between the calculated fault reactance and actual line reactance. This method overcomes the requirement of uniformed parameters in traditional impedance method, and the simulation results show that it can fault location accurately in distribution network.

Keywords—distribution network, fault location, apparent impedance method, binary search

I. INTRODUCTION

In the power system, distribution network is a power transmission network directly connected with users. Therefore, the safe and stable operation of the distribution network is directly related to the power reliability of the user side. According to statistics, the proportion of outages caused by distribution network faults is as high as 95% [1]. For the distribution network, more than 80% of the faults are single-phase grounding faults [2-3]. Therefore, timely detection and isolation of faults is important for the reliability of users' power consumption. Compared with the transmission network, the structure of distribution network is more complex and the types of conductors are heterogeneous. At the same time, in the urban power grid, there are many sub-branches in the transmission lines which are composed of overhead lines and cables [4]. As a result, the difficulty of fault location in distribution network is greatly increased. Many methods based on uniform distribution parameters are not suitable for fault location in distribution network. Meanwhile, the fault indicators based methods cannot apply in distribution network because that not all nodes installed FTU and other automation monitoring terminals [5].

Currently, fault location methods of distribution network can be divided into two categories: one is the methods based on fault signal, the other is the methods based on active injection signal[6]. Due to the low coverage of distribution automation in China, fault location methods based on distribution automation monitoring terminals and fault indicators cannot be adopted in current environment. However, the fault signal based methods, such as impedance method, is not suitable for the non-uniform line parameters in distribution network. With the power grid company attaching more importance to power quality, power quality monitoring terminals are generally installed at the substation of 10 kV distribution network. A large number of steady-state and transient data can be provided by power

quality monitoring terminals[7]. Making full use of these data can provide high-precision data information for distribution network fault location under the condition of insufficient automation.

Based on the automation level of distribution network and the structure characteristics of distribution network, this paper proposed an improved apparent impedance method for fault location in distribution network. Under the condition of low coverage of distribution automation in China, it can overcome the shortcomings of traditional impedance method, which has the requirement for uniform line parameters. The proposed method can fault location accurately, so it can ensure the safety of distribution network and the reliability of users' electricity consumption.

II. BASIC PRINCIPLE OF FAULT LOCATION

A. Analysis of Fault Voltage and Current in Single Phase Grounding Fault

Impedance method belongs to the single-terminal fault location algorithm, which can be realized only by the voltage and current data at the substation of the distribution network. At the same time, the accurate parameters of the distribution transmission line are required in the method. Obviously, it has good economy and it adapts to the changes of the operation mode of the distribution network[8]. Fig. 1 is a schematic diagram of a single-phase-ground fault occurring at the distribution line. And a monitoring device, such as power quality monitoring device, is installed at the substation X.

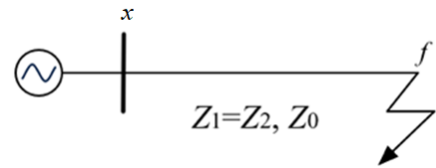


Figure.1. The diagram of a single-phase-to-ground fault

Assuming that the fault phase is A, the sequence components of the voltage and current at the fault point are shown as in Eq.(1)-Eq.(3). According to the marginal condition of the short circuit fault, the Eq.(4) can be obtained.

$$V_{1f} = V_1 - I_1 Z_1 \quad (1)$$

$$V_{2f} = V_2 - I_2 Z_2 \quad (2)$$

$$V_{0f} = V_0 - I_0 Z_0 \quad (3)$$

At the same time:

$$V_{1f} + V_{2f} + V_{0f} = 3I_{0f} R_f \quad (4)$$

Where,

V_i is the sequence component of the voltage at node X;

I_i is sequence component of the current at node X;

Z_i is the sequence impedance of fault circuit;

V_{if} is the sequence component of the voltage at the fault point.

Assuming that the positive sequence impedance and the negative sequence impedance of the transmission line are same, and Eq.(5) can be obtained from Eq.(1) to Eq. (4).

$$V_{af} = V_a - (I_1 + I_2)Z_1 + 3I_0Z_0 \quad (5)$$

Eq.(5) can also be written as follows:

$$V_a = (I_a + kI_0)Z_1 + 3I_0R_f \quad (6)$$

Where,

$$k = \frac{Z_0 - Z_1}{Z_1}$$

B. Fault distance calculation based on apparent impedance method

The apparent impedance is defined as the ratio of voltage and current at the selected point [9], the calculation formula is shown in Eq.(7).

$$Z_{app} = \frac{V_{select}}{I_{select}} = R_{app} + jX_{app} \quad (7)$$

Where, R_{app} is apparent impedance and X_{app} is apparent reactance.

Then, Eq. (6) can be written as follows:

$$\frac{V_a}{I_a + kI_0} = Z_1 + \frac{3I_0R_f}{I_a + kI_0} \quad (8)$$

According to the definition of apparent impedance, it can be defined as Eq.(9) and Eq.(10).

$$V_{select} = V_a \quad (9)$$

$$I_{select} = I_a + kI_0 \quad (10)$$

Combining the Eq.(7)-Eq.(10), we can get that:

$$Z_{app} = Dz_1 + \frac{3I_0R_f}{I_a + kI_0} \quad (11)$$

Where, z_1 is the unit positive sequence impedance of the transmission lines from the fault point to the monitoring point; D is the distance from the fault point to the monitoring point and R_f is the fault resistance.

In Eq. (11), the fault resistance R_f is unknown. In order to eliminate the fault resistance, Eq.(11) can be written as the real part and the imaginary part, respectively.

$$\begin{bmatrix} R_{app} \\ X_{app} \end{bmatrix} = \begin{bmatrix} r_1 & \frac{I_d I_{s1} + I_q I_{s2}}{I_{sm}^2} \\ x_1 & \frac{I_d I_{s2} + I_q I_{s1}}{I_{sm}^2} \end{bmatrix} \begin{bmatrix} D \\ R_f \end{bmatrix} \quad (12)$$

In the Eq.(12), r_1 is the unit positive sequence resistance of the line from the fault point to the monitoring point;

x_1 is the unit positive sequence reactance of the line from the fault point to the monitoring point;

I_{s1} is the real part of $I_a + kI_0$;

I_{s2} is the imaginary part of $I_a + kI_0$;

I_d is the real part of $3I_0$;

I_q is the imaginary part of $3I_0$;

$$I_{sm}^2 = I_{s1}^2 + I_{s2}^2.$$

By eliminating the unknown R_f , the distance D from the fault point to the monitoring point can be obtained in Eq.(13).

$$D = \frac{R_{app}M - X_{app}L}{r_1M - x_1L} \quad (13)$$

$$\text{Where, } L = \frac{I_d I_{s1} + I_q I_{s2}}{I_{sm}^2},$$

$$M = \frac{-I_d I_{s2} + I_q I_{s1}}{I_{sm}^2}.$$

III. IMPROVED APPARENT IMPEDANCE METHOD FOR FAULT LOCATION

As mentioned before, the assumption of the apparent impedance method is that the line parameters are uniformly distributed. In fact, the distribution network structure is complex. It includes various types of overhead lines and cables. Therefore, the line parameters are also different. If the method based traditional apparent impedance is employed to calculate the fault position, it will bring greater errors. Therefore, an improved positive sequence impedance method is proposed in this paper, which is suitable for the actual situation of distribution network in China. Compared with the traditional impedance method, this proposed method considers the non-uniform distribution of line parameters. It can improve the accuracy of fault location by taking into account the parameters of different types of conductors.

In Eq.(13), x_1 is multiplied in both sides of the equation at the same time:

$$Dx_1 = \frac{R_{app}M - X_{app}L}{\frac{r_1}{x_1}M - L} \quad (14)$$

Where, Dx_1 is the total positive sequence reactance from the monitoring point to the fault point, which is defined as X_{1_fault} .

In the upper denominator of Eq.(14), $\frac{r_1}{x_1}$ is the ratio of the average positive sequence resistance and the positive sequence reactance from the fault point to the monitoring point. When it multiplied by the length of the bus path, the

ratio of the total positive sequence resistance and the total positive sequence reactance can be obtained in Eq. (15):

$$X_{1_fault} = \frac{R_{app}M - X_{app}L}{\frac{R_1}{X_1}M - L} \quad (15)$$

Where, R_1 is the total positive sequence resistance from the monitoring point to the fault point. X_1 is the total positive sequence reactance from the monitoring point to the fault point.

Because the parameters of each conductor are different, it is impossible to calculate the fault distance directly. However, by assuming the position of the fault point, the difference between the calculated positive sequence reactance and the actual reactance can be compared to judge the relationship between the hypothetical fault point and the actual fault point position. Therefore, the position of the hypothetical fault point can be further adjusted. When the difference between them is less than the expected value, the assumed fault location is judged as the actual fault location.

Figure 2 is a simplified diagram of the distribution network, and the actual fault occurs at f . In order to search of the actual fault point, this paper adopts binary search method[10-11]. It means that the total actual positive sequence reactance X_{1_total} is calculated at each conductor segment and the X_{1_fault} is also calculated, which is the reactance value at assuming fault point. By comparing the difference between the two reactance value, the fault segment can be determined. And then gradually narrow the scope of the actual fault point by dichotomy until the difference between the two reactance value is less than the expected value.

According to Fig. 2, the short-circuit fault actually occurs at f , and the X_{1_total} and X_{1_fault} at each conductor connection are calculated respectively. Among them, X_{1_total} is the sum of the total positive sequence reactance of each section of conductor, and X_{1_fault} is calculated by Eq.(15).

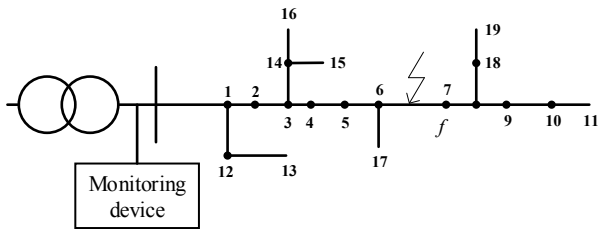


Figure .2. The diagram of a single phase to ground in distribution network

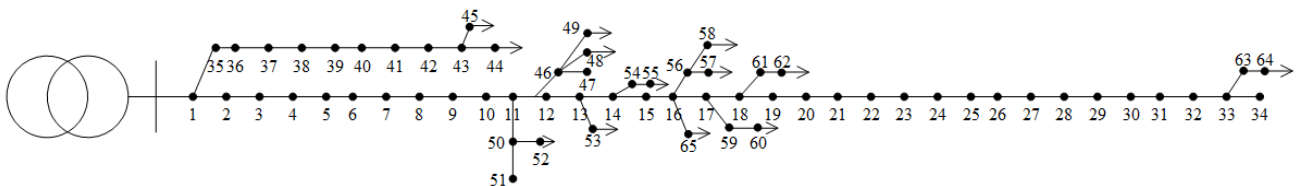


Figure.4. The topology of distribution network

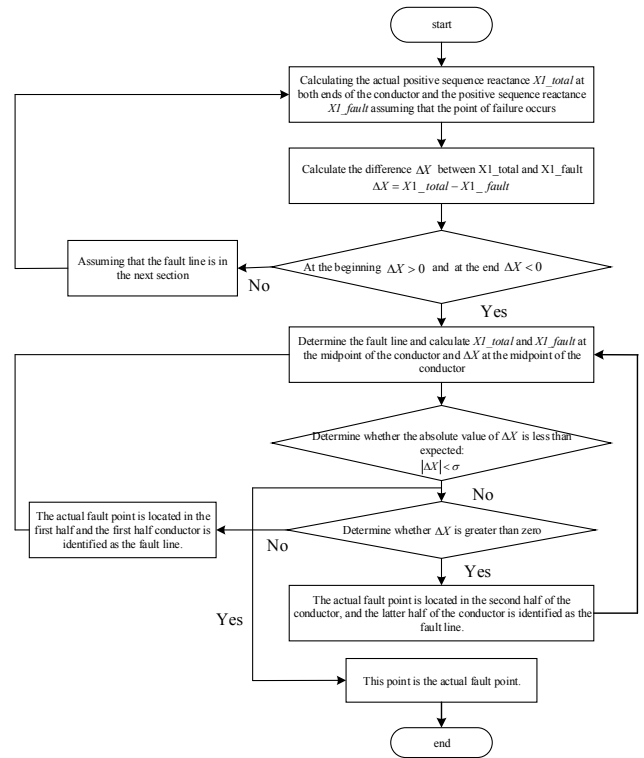


Figure. 3. The flowchart of fault location algorithm

According to the actual position of the fault point, the calculated X_{1_fault} at each node before the fault point is larger than the actual total positive sequence reactance at the node, while the X_{1_fault} at the rear of the actual fault point is smaller than the actual total positive sequence reactance. According to this condition, the conductor segment where the actual fault point can be determined by the trends of the difference between X_{1_total} and X_{1_fault} . The difference between X_{1_total} and X_{1_fault} is denoted as ΔX :

$$\Delta X = X_{1_total} - X_{1_fault} \quad (16)$$

After determining the fault segment, the X_{1_total} , X_{1_fault} and ΔX between them at the midpoint of the conductor are calculated. If ΔX is less than zero, it means that the total positive sequence reactance X_{1_fault} at the fault point is greater than the actual total positive sequence reactance. Therefore, the fault point is located in the latter half of the conductor. Otherwise, The barrier point is located in the first half of the conductor. This process is continued until ΔX is less than the threshold σ . When the difference between the calculated positive sequence reactance and the actual positive sequence reactance is very small, the calculated position is the fault location. The flow chart of the improved apparent impedance method is shown in Fig. 3.

IV. SIMULATION VERIFICATION

OpenDSS is a distribution network modeling and simulation software developed by American Academy of Sciences. It is very convenient for distribution network modeling and fault simulation. At the same time, OpenDSS provides variety interfaces, which are used to call OpenDSS model and parameters through different software. In this paper, OpenDSS is used to model the distribution network in a certain area. And the distribution network model of OpenDSS is invoked by MATLAB. A lot of simulation result verified that the proposed fault location method can fault position accurately.

A. Distribution Network Modeling

OpenDSS provides variety power components of distribution network. According to the collected line information, the actual distribution line model is established in OpenDSS. Figure 4 shows the distribution network topology of the area.

B. Fault simulation

The OpenDSS interface is invoked in Matlab to set short-circuit faults at different conductors artificially. And the power flow data after the fault are obtained by OpenDSS. The improved apparent impedance method is used to locate the fault point. Table 1 is the results of fault location by simulation.

TABLE I. FAULT LOCATION RESULT OF IMPROVED APPARENT IMPEDANCE METHOD

Actual distance(km)	Calculating distance(km)	Calculation error(km)	Error rate%
0.828	0.851	0.023	2.78
1.735	1.681	-0.054	-3.11
2.698	2.586	-0.112	-4.15
3.407	3.264	-0.143	-4.20
4.361	4.306	-0.055	-1.25

From the calculation and simulation results, it can find that the error tends to increase gradually with the increase of the distance between the fault point and the monitoring device. However, for the length of the distribution network line, the overall error is within acceptable range.

C. Analysis of the Impact of Fault Location Accuracy

a) Effect of Fault Resistance

The results of fault location in Table 1 are calculated when the fault resistance R_f is 0.1. In order to analysis the influence of fault resistance on the fault location, the fault resistance is set to 1, and the results are shown in Table 2.

TABLE II. FAULT LOCATION RESULT WITH $R_f=1$

Actual distance(km)	Calculating distance(km)	Calculation error(km)	Error rate%
0.828	0.850	0.023	2.78
1.735	1.685	-0.050	-2.88
2.698	2.586	-0.112	-4.15
3.407	3.256	-0.151	-4.43
4.361	4.297	-0.064	-1.47

In the Table 2, it can be seen that the fault resistance will have a certain impact on results of the fault location, but the impact is small.

b) Effect of Line Loads

The fault of distribution network with low load rate has been tested in Table 3. The results show that the fault error is less than 5%.

TABLE III. FAULT LOCATION RESULT OF WITH LOW LOAD

Actual distance(km)	Calculating distance(km)	Calculation error(km)	Error rate%
0.828	0.849	0.021	2.53
1.735	1.687	-0.048	-2.77
2.698	2.576	-0.112	-4.51
3.407	3.254	-0.143	-4.49
4.361	4.300	-0.055	-1.40

V. CONCLUSION

Due to the complex structure and conductors of distribution network, accurate fault location is very different. Furthermore, it is meaningful for develop a single-ended fault location method for distribution network in China with low automation level. The improved apparent impedance method proposed in this paper takes into account the actual situation of the distribution network currently. Based on the mixed simulation of OpenDSS and MATLAB, a lot of simulation tests which are in the actual distribution network in certain area verify the reliability of the proposed method. At the same time, this paper also analyzed the factors affecting the accuracy of fault location. The results shows that the fault resistance and line load will have a certain impact on the accuracy of fault location, but the fault location error is within acceptable range.

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